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Analog Design Experiments With AI—Part 2

In the previous article [1], we assessed the analytical abilities of ChatGPT for basic circuits. In this article, we extend our effort toward circuit *design* by asking artificial intelligence (AI) to predict performance trends. As in the previous article, we give a score from zero to four to AI's answers.

The Analog Design Pyramid

To create a cohesive framework for our studies, we first ponder a general question: How can AI become a competent analog designer? We postulate that AI must climb a pyramid of five levels, as depicted in Figure 1. At the foundation, AI must analyze circuits correctly. As explained below, this ability appears to be maturing in some AI tools, especially if it simply requires writing and solving equations. The second level relates to intuition: Can AI understand a structure well enough to predict its behavior as we vary certain parameters? Possessed by good analog engineers, this key skill guides the design process, distinguishing those engineers from "SPICE monkeys."

At the third level, AI must be able to design a given circuit topology according to certain specifications. We see below that the task of training AI for this purpose presents formidable challenges. The fourth level extends this ability to multiple topologies for the same function, seeking a global optimum.

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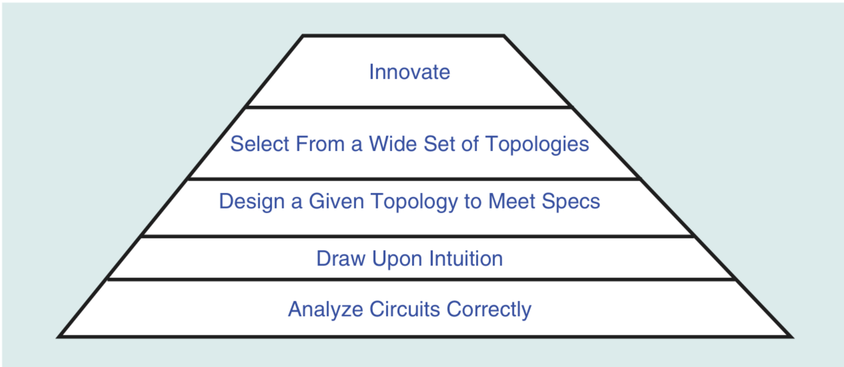

FIGURE 1: A pyramid of five levels for analog design. XOR: exclusive or.

The fifth level of the pyramid calls for innovation in circuits, in a manner similar to what humans have done for a century.

The Vast Universe of Analog Design

Training AI to climb the pyramid of Figure 1 faces interesting issues. Besides the question of how intuition should be taught, we should recognize the enormity of analog design training. In the digital world, we encounter about a dozen basic cells (Figure 2), and we optimize them in terms of their speed–power tradeoffs. In the analog world, on the other

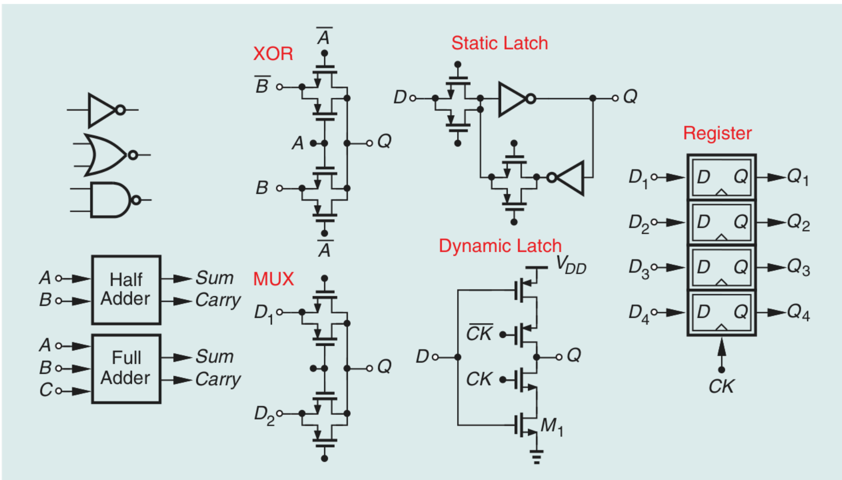

FIGURE 2: Basic digital building blocks. DMUX: demultiplexer; CS: common-source; CG: common-gate; CTLE: continuous-time linear equalizer; SST: source-series termination; DTC: digital-to-time converter.

hand, we encounter about 50 distinct building blocks (Figure 3), while facing tradeoffs among speed, power, noise, linearity, voltage headroom, and supply rejection. The third level in our pyramid must therefore deal with all of these issues.

Training AI for analog design proves even more challenging if we observe that each of the building blocks in Figure 3 can assume different implementations. As an example, Figure 4 shows a number of LNA structures that find usage in different applications. At the fourth level, AI must be trained to design all of these LNAs. Extending this point to the cells in Figure 3, we can estimate hundreds of circuit topologies that AI must learn to handle.

Images Versus Netlists

In our interaction with AI tools, we can submit images or netlists for circuit structures, the latter causing less confusion. In some cases, however, providing netlists proves impractical. For example, suppose, as part of training, AI is to read articles or book chapters containing tens of circuit diagrams. It would be rather cumbersome to convert all of such images to netlists. In other words, we expect, at a minimum, that AI correctly recognizes the devices constituting a given circuit.

In the previous article [1], we observed that ChatGPT found it difficult to decipher some circuit

CS, CG, Followers	Current Steering DACs	CTLEs
Diff. Pairs	Sampling Circuits	Phase Interpolators
Cascodes	Frequency Multipliers	Duty Cycle Correction
Operational Amplifiers	DCOs	SST Drivers
Feedback Circuits	VGAs	PAM4 Drivers
Bandgaps	$\Delta\Sigma$ Modulators	DTCs
DC–DC Converters	Digital PLLs	TDCs
Ring VCOs	RF Synthesizers	Crystal Oscillators
LC VCOs	CDRs	Voltage Multipliers
Frequency Dividers	FFEs	LNAs
Phase Detectors	DFEs	Downconversion Mixers
Charge Pumps	MUX/DMUX	Upconversion Mixers
PLLs	CT Integrators	Baseband Filters
DLLs	DT Integrators	PAs
Comparators	Bidirectional Hybrids	Polyphase Filters
Capacitor DACs	TIAs	Quadrature Generation
Resistor Ladder DACs	Offset Cancellation	Quadrature Correction

FIGURE 3: Basic analog building blocks.

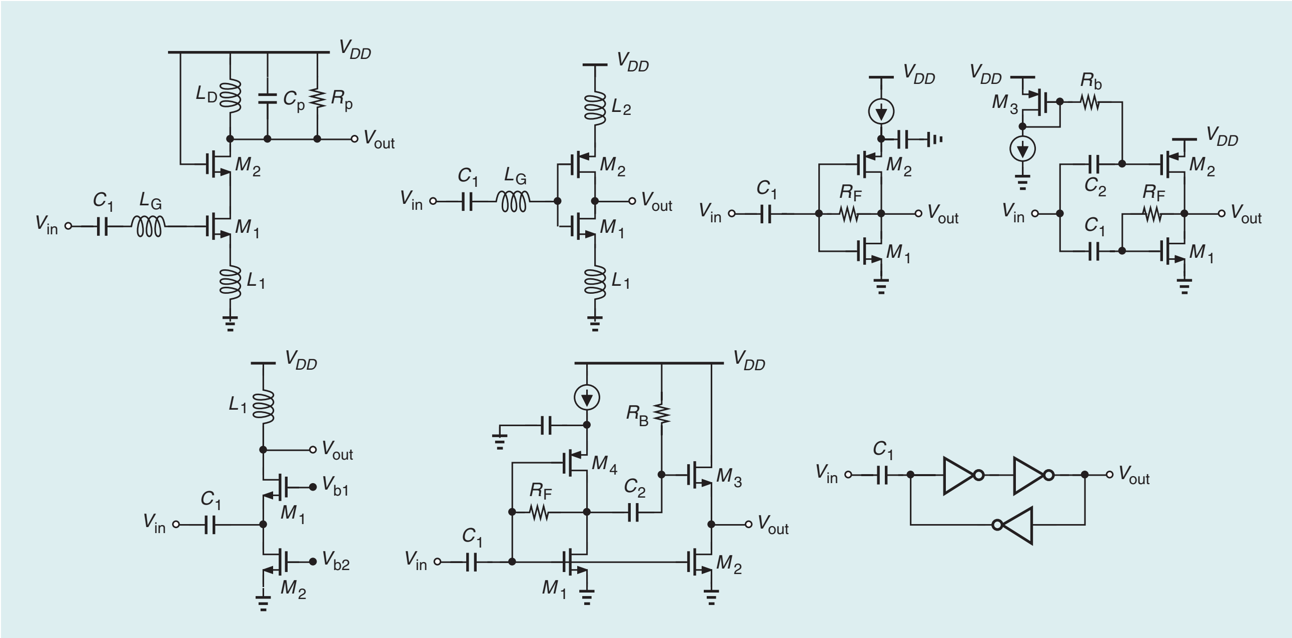

FIGURE 4: Typical LNA topologies.

diagrams. Fortunately, other tools, such as Gemini 3 or Gemini 3.1 Pro, exhibit much greater proficiency in reading images. We therefore continue our endeavor with Gemini 3.1 Pro.

Ring Oscillator Examples

Let us consider some questions that a designer faces with respect to ring oscillators.

Q1: Does the circuit in Figure 5 fail to oscillate if the three capacitors become arbitrarily large?

Gemini 3.1 Pro: No, it does not. The circuit simply operates at a lower frequency. (A small-signal analysis is also provided.)

Comments: This answer is correct. The small-signal analysis and even the notation used by the tool closely resemble those in two of my books [2], [3]. Score: 4

Q2: What happens to the phase noise if the three capacitors in Figure 5 are doubled?

Gemini 3.1 Pro: The phase noise falls by a factor of four. (An analysis based on the edge rate is also provided.)

Comments: The answer is correct. We can draw upon this equation for phase noise due to white noise:

$$S_{\phi n}(\Delta f) = \left(\frac{f_0}{\Delta f}\right)^2\left[\frac{1}{2I_D^2}(S_{I,\mathrm{NMOS}}+S_{I,\mathrm{PMOS}})+\frac{2kT}{I_D V_{DD}}\right] \quad (1)$$

where f_0 is the oscillation frequency and the other parameters are described in [4] and independent of

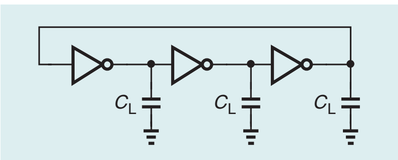

FIGURE 5: A simple ring oscillator.

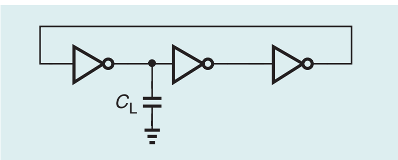

FIGURE 6: A ring oscillator loaded by a capacitor.

C_L. Since f_0 is halved, $S_{\phi n}(f)$ drops by a factor of four. Score: 4

Q3: Does the circuit in Figure 6 fail to oscillate if C_L becomes arbitrarily large?

Gemini 3.1 Pro: Yes, it does. (A small-signal analysis is also provided.)

Comments: The answer is correct. For a sufficiently large C_L, the resulting dominant pole introduces so much gain roll-off that the loop gain falls below unity at the phase crossover frequency. Score: 4

Q4: If we double the widths of the NMOS and PMOS devices in Figure 7, what happens to the phase noise?

Gemini 3.1 Pro: The phase noise drops by 3 dB.

Comments: The answer is correct. This trend can be proved using a "linear scaling" argument or by viewing the result as two rings placed in parallel. Score: 4

Q5: Determine the small-signal resistance seen looking into the supply node of the ring oscillator in Figure 8.

Gemini 3.1 Pro: We have

$$R_X \approx \frac{1}{3}\left(\frac{1}{g_{m1}}+\frac{1}{g_{m2}}\right) \quad (2)$$

because each inverter behaves as two diode-connected devices in series.

Comments: The answer is incorrect. This question is not just an aca-

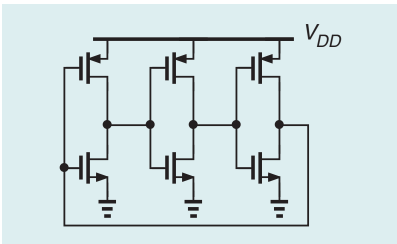

FIGURE 7: A ring oscillator shown at the transistor level.

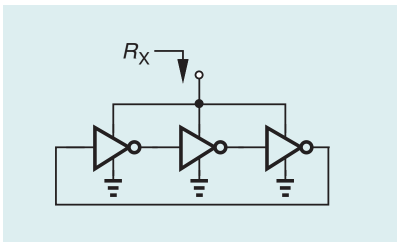

FIGURE 8: The impedance seen in the supply node of a ring oscillator.

demic curiosity; rather, the value of R_X determines the roll-off frequency if a capacitor is attached to this node [5]. As explained in [5], the charge and discharge of the capacitance, C_L, at each node within the loop create an equivalent resistance of $1/(f_0 C_L)$, where f_0 denotes the oscillation frequency, yielding

$$R_X \approx \frac{1}{3 f_0 C_L}. \quad (3)$$

Alternatively, we can express the power consumption as $P = 3 f_0 C_L V_{DD}^2 = V_{DD} I_{DD}$, where I_{DD} is the average supply current. That is, $\partial I_{DD}/\partial V_{DD} = 3 f_0 C_L$. Gemini ignores oscillation here. Score: 1

Q6: If the circuit oscillates, does your answer change?

Gemini 3.1 Pro: Yes, it does. The average resistance is now much higher because the path from V_{DD} to ground through the transistors is off most of the time.

Comments: The answer is still incorrect. Writing $f_0 = 1/(6T_D)$, where T_D is the gate delay, we have

$$R_X \approx \frac{2T_D}{C_L}. \quad (4)$$

We can approximate T_D as $R_{eq} C_L$, where R_{eq} is on the order of the on-resistance of the MOSFETs and, hence, comparable to their inverse transconductances. Thus, R_X is *not* much higher than that in (2). Score: 1

Comparator Examples

We ask Gemini 3.1 Pro to read an article on the StrongARM comparator [6]. We then pose questions.

Q7: In the circuit of Figure 9, what happens to the initial voltage gain as the width of M_7 increases?

Gemini 3.1 Pro: As this width increases, the gain drops. Equation (1) in the article,

$$A_v \approx \frac{g_{m1,2} V_{THN}}{I_{CM}} \quad (5)$$

suggests that a greater I_{CM} $(= I_{D7})$ lowers A_v because $g_{m1,2} \propto \sqrt{I_{CM}}$.

Comments: The answer is correct. Score: 4

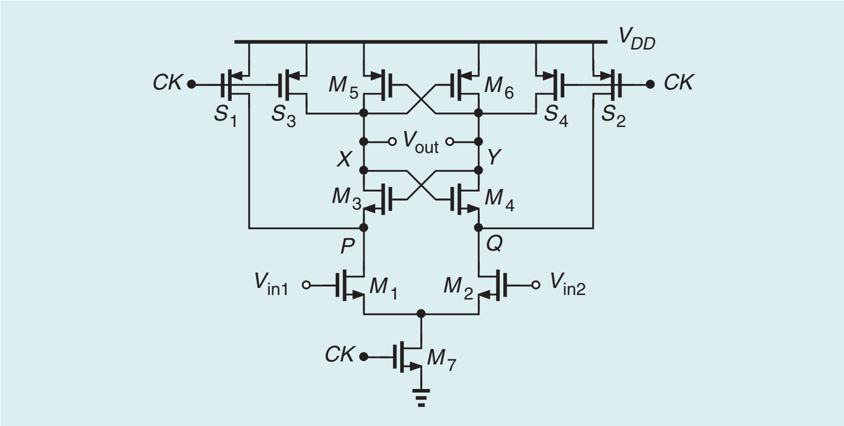

FIGURE 9: The StrongARM comparator.

Q8: In Figure 9, what happens to the initial voltage gain if we increase the capacitance at nodes *P* and *Q*?

Gemini 3.1 Pro: The gain remains constant because (5) does not depend on this capacitance.

Comments: The answer is correct. Score: 4

Q9: If we increase the widths of M_5 and M_6 in Figure 9, does the speed improve or degrade?

Gemini 3.1 Pro: It degrades because their transconductance increases by a smaller factor than do their capacitances.

Comments: The answer is incorrect. Since the other capacitances at *X* and *Y* dominate initially, as we increase the widths of M_5 and M_6, the speed improves up to some point. Score: 1

Frequency Divider Examples

In this section, we ask AI to study a number of divide-by-two circuits.

Q10: In the circuit of Figure 10, does the speed improve or degrade if we increase the widths of the clocked transistors?

Gemini 3.1 Pro: The speed degrades because the data-driven devices are the bottleneck.

Comments: The answer is not quite correct, and for the same reason. If the data-driven devices are the bottleneck, as we increase the widths of the clocked transistors, the speed improves initially even though these devices introduce capacitances in the data path. Score: 2

Q11: Does the structure in Figure 10 provide quadrature outputs?

Gemini 3.1 Pro: Yes, it does. The output at *Y* or *Z* is in quadrature with respect to that at *X*.

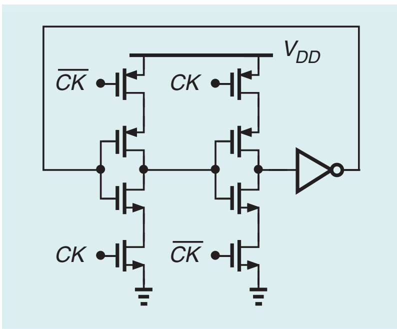

FIGURE 11: Another C²MOS divide-by-two circuit.

Comments: The answer is not quite correct. The delay through Inv_3 causes a skew in *Z*. Also, it is difficult to ensure equal load capacitances seen by the two latches. For example, the gate drain capacitances within Inv_3 experience Miller multiplication, but those within the secondary latch do not. Score: 2

Q12: Repeat question 10 for the topology in Figure 11.

Gemini 3.1 Pro: The speed degrades, and charge sharing leads to catastrophic results.

Comments: The answer is incorrect. In fact, the speed improves considerably because the clocked devices do not interfere with the data path much. Charge sharing does occur, but the degraded swings are acceptable up to some point. Gemini receives a score of 1 here for discouraging the designer from exploring this useful circuit.

Q13: Analyze the arrangement in Figure 12(a).

Gemini 3.1 Pro: This is a divide-by-two circuit using dynamic latches.

Comments: The answer is correct. Score: 4

Q14: How does the red inverter in Figure 12(b) affect the performance?

Gemini 3.1 Pro: The inverter acts as a keeper, forming a cross-coupled latch with the first inverter. This removes the minimum clock frequency requirement (due to leakage).

Comments: The answer is incorrect. The inverter creates a feedforward path around the first latch, improving the divider speed. Since this unclocked branch begins to overwhelm the main path at low clock frequencies,

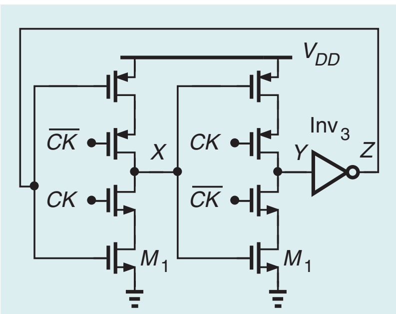

FIGURE 10: A C²MOS divide-by-two circuit.

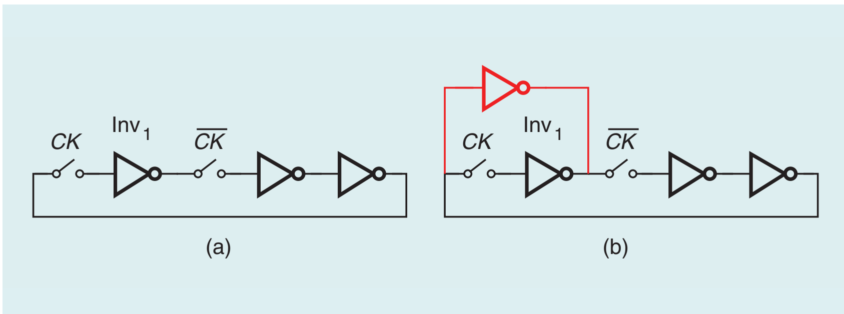

FIGURE 12: (a) A divide-by-2 circuit and (b) the structure with feedforward added.

feedforward, in fact, imposes a higher bound on the lowest clock rate that the circuit can handle. Score: 0

It appears that, in its attempt to map this topology to any other one that it knows, Gemini decides that the red inverter is cross-coupled with Inv_1.

Q15: But the input of the red inverter is not tied to the output of Inv_1.

Gemini 3.1 Pro: I apologize. This creates a problematic configuration. There must be a mistake in this drawing.

Comments: Even facing feedback from the user, AI simply dismisses the circuit. Score: 0

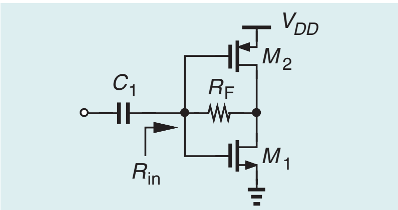

FIGURE 13: A simple LNA.

Amplifier Examples

Q16: How do we optimize the circuit of Figure 13 for the NF if R_{in} must remain equal to 50 Ω?

Gemini 3.1 Pro: The NF is given by

$$F \approx 1 + \frac{R_s}{R_F} + \gamma \tag{6}$$

and the input resistance by

$$R_{\text{in}} = \frac{R_F + R_L}{1 + g_{m,\text{tot}} R_L} \tag{7}$$

where R_L is the load resistance. We maximize R_L by choosing long-channel devices so that $R_{\text{in}} \to 1/g_{m,\text{tot}}$. The lowest NF is given by $1+\gamma$ if $R_{\text{in}} = R_s = 1/g_{m,\text{tot}}$ and R_F is large enough.

Comments: The answer is incorrect. As explained in [7], the presence of channel length modulation rewrites the relation between R_{in} and the NF, making it possible to achieve $\text{NF} < 1+\gamma$. Gemini, on the other hand, recommends to raise the output resistance so as to approach the lower bound of $\text{NF} = 1+\gamma$. Score: 1

Q17: Compute the input impedance of the circuit in Figure 14(a).

Gemini 3.1 Pro: This is a shunt–shunt feedback amplifier.

$$R_{\text{in}} = \frac{R_F}{1 + |A_{\text{vol}}|} \tag{8}$$

where

$$A_{\text{vol}} = -\frac{1}{2} g_{m1} g_{m2,3} R_{D1} (R_{D2} \,\|\, R_F). \tag{9}$$

Comments: The answer is not quite correct. Gemini attempts to solve the circuit through the use of feedback techniques. However, if we simply model the amplifier as illustrated in Figure 14(b), we have, from Miller's theorem,

$$R_{\text{in}} = \frac{R_F + R_{D2}}{1 + A_0} \tag{10}$$

where A_0 is the unloaded gain and equal to

$$A_0 = \frac{1}{2} g_{m1} g_{m2,3} R_{D1} R_{D2}. \tag{11}$$

This result is exact. Score: 2

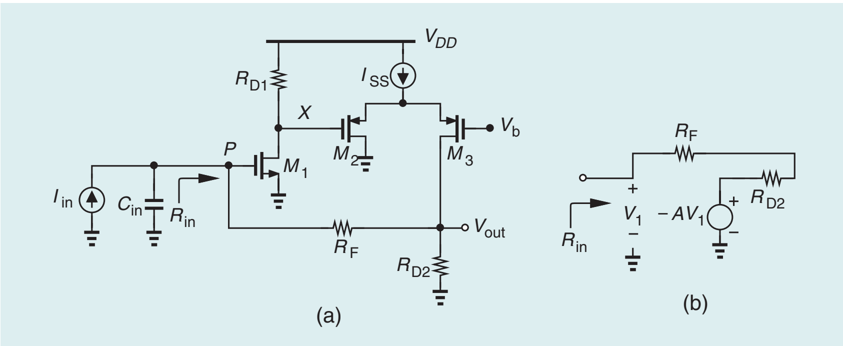

FIGURE 14: (a) A TIA and (b) its simplified model.

LC Oscillator Examples

Q18: If the LDO regulator in Figure 15 generates thermal noise with spectrum S_{th}, how do we compute the VCO output phase noise?

Gemini 3.1 Pro: We need to find the VCO pushing gain, defined as

$$K_{\text{push}} = \frac{\partial f_{\text{osc}}}{\partial V_{\text{out}}} \tag{12}$$

typically found by SPICE simulations. Then,

$$L(\Delta f) = 10 \log\left(\frac{K_{\text{push}}^2 \cdot S_{th}}{2\Delta f^2}\right). \tag{13}$$

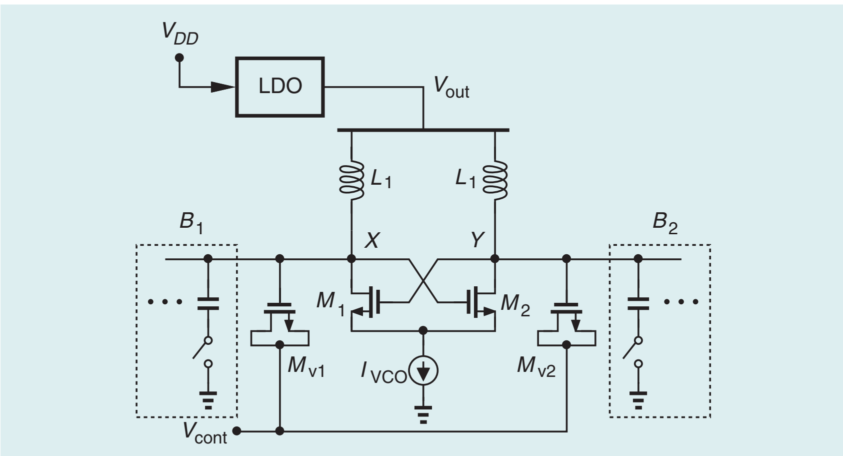

FIGURE 15: An *LC* oscillator fed from an LDO.

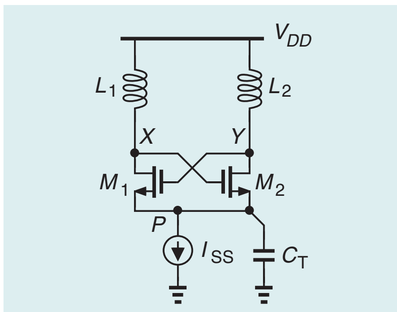

FIGURE 16: An *LC* oscillator with a capacitor added to its tail node.

Comments: The answer is correct. However, Gemini fails to see that $K_{\text{push}} \approx K_{\text{VCO}}$: a perturbation on the VCO supply appears directly on the varactors. Knowing this relation, a designer would opt for a lower K_{VCO}. Score: 3

Q19: We add C_T to the tail node, as in Figure 16. What happens to the phase noise?

Gemini 3.1 Pro: It generally lowers the phase noise by shunting the I_{SS} noise at $2\omega_0$ and by suppressing the $2\omega_0$ voltage ripple at the tail node, which could push the transistors into the triode region. The capacitor pins P to ground and ensures that M_1 and M_2 are in saturation.

Comments: The answer is mostly incorrect. As C_T increases, initially, the flicker noise of M_1 and M_2 is upconverted to phase noise (unless they are perfect square law devices [8]). However, if C_T reaches a certain range, M_1 and M_2 experience class C switching, delivering peak currents to the tanks that are greater than I_{SS} and lowering the phase noise [8]. In this mode, the large currents can, in fact, drive M_1 and M_2 into the triode region. As C_T becomes large enough, the circuit exhibits low-frequency instability. Score: 1

Q20: Estimate the oscillation frequency of the circuit in Figure 17.

Gemini 3.1 Pro: If the total capacitance at each node is C_{node}, then, $\omega_{\text{osc}} = \omega_0 + \Delta\omega$, where

$$\omega_0 = \frac{1}{\sqrt{LC_{\text{node}}}} \tag{14}$$

and

$$\Delta\omega = \frac{g_{mc}}{2C_{\text{node}}}. \tag{15}$$

Here, g_{mc} denotes the transconductance of the coupling pairs.

Comments: The answer is incorrect. The shift from resonance is given by

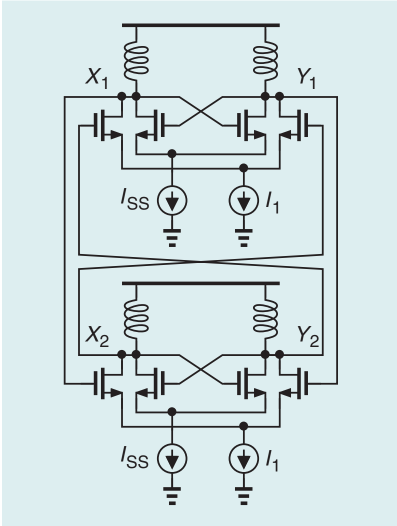

FIGURE 17: A quadrature oscillator.

Gemini 3.1 Pro Results

Q1	4	Q11	2
Q2	4	Q12	1
Q3	4	Q13	4
Q4	4	Q14	0
Q5	1	Q15	0
Q6	1	Q16	1
Q7	4	Q17	2
Q8	4	Q18	3
Q9	1	Q19	1
Q10	2	Q20	1

Total: 44/80
(55%)

FIGURE 18: The results of this assessment.

$$\Delta\omega = \alpha\frac{\omega_0}{2Q} \tag{16}$$

where α is the coupling coefficient. In a typical design, the transistors experience nearly complete switching, yielding $\alpha = I_1/I_{SS}$, and hence,

$$\Delta\omega = \frac{I_1}{I_{SS}}\cdot\frac{\omega_0}{2Q}. \tag{17}$$

Even if we assume small-signal operation, we have $\alpha = g_{mc}/g_{m1}$, where g_{m1} is the transconductance of the cross-coupled transistors. It follows that

$$\Delta\omega = \frac{g_{mc}}{g_{m1}}\cdot\frac{\omega_0}{2Q} \tag{18}$$

$$= \frac{g_{mc}}{g_{m1}}\cdot\frac{1}{2C_{\text{node}}R_p}. \tag{19}$$

It appears that Gemini assumes $g_{m1}R_p = 1$; i.e., the circuit resides at the edge of oscillation. Score: 1

Conclusion

Figure 18 compiles the scores that Gemini 3.1 Pro received in this assessment. With a total of 55%, this AI tool exhibits some potential for analog design but requires more extensive training. Of course, how AI will climb to the top of our pyramid remains to be seen.

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